

# Summer Internship Project Report

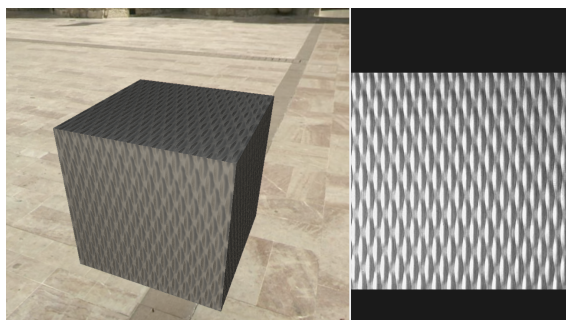
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June - August 2026

**AI Disclosure:** Claude AI and ChatGPT were both used in a research context to teach software (Blender), generate code (Overleaf LaTeX), and answer general questions. Generative AI was not used to write any part of this report.

# 1 Projects

## 1.1 Awesome Bump



(a) 5WL Awesome Bump base map.



(b) 5WL bump map projected onto a custom polished-copper solid body created in Inventor.

Figure 1: The 5WL Awesome Bump base map and its application to the custom solid body.

Toward the beginning of my internship, I did some general research on software that could help me achieve the proposed goal of 3D modeling a custom pattern. I came across Awesome Bump from a GitHub repository and learned the basics through YouTube videos. However, I quickly learned that Awesome Bump was better at projecting texture images onto a given surface rather than actually creating the 3D aspect. I applied a couple of different texture samples from the Rigidized Metals website and scaled them to fit the object. Figure 1a shows the 5WL pattern made into a bump map prior to export in Inventor. I also wrote a manual for creating a bump visualization so that anyone can create their own. Although this project did not end up being particularly useful in modeling a 3D textured pattern, it is useful for creating a mock-up of a customer's desired product for a realistic visualization, as demonstrated in Figure 1b.

## 1.2 Laser Scanning and Data Collection

Another skill set I acquired from the beginning of my internship was using the optical profilometer to scan sheet metal samples and collect data. My mentor and I began by selecting several popular patterned (5WL, 6WL, 1RL) 6" x 6" sample squares in various gauges (0.024", 0.029", 0.060") from the sample department. I learned to use a point micrometer to measure the base thickness of a sample and metal thickness on a character. Then I learned to use a disk micrometer to measure the pattern depth at various locations on the sample and averaged them to get an accurate estimate. With the VR-6000 I generated profile

graphs and took a depth measurement using the highest and lowest points on the graph. I also measured the surface area of a selection of three to five characters to then subtract the surface area of a flat sheet with the same dimension to calculate the additional surface area the pattern adds to the sheet. The end goal is to have a catalog of sample reports of all the patterns on the different gauges that Rigidized Metals offers for marketing uses. Since there are hundreds of different combinations of patterns and gauges, the catalog has not been completed by the end of my internship; in lieu of this, I worked alongside my mentor to write a data collection instruction manual using the VR-6000. This manual guides a beginner user through taking measurements by hand using point and disk micrometers, scanning the sheet metal, adjusting scale parameters, collecting the appropriate data, and generating a report.



Figure 2: Keyence VR-6000 Optical Profilometer

## 1.3 Custom Roll Creation

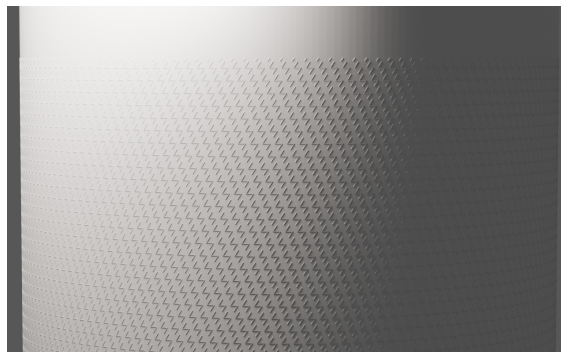
### 1.3.1 Blender

To learn Blender, I watched many YouTube videos and read online forums over the course of several weeks. I started by utilizing the Sculpting tab to pull and push material from spheres and planes to create a bumpy texture. Finding this method too imprecise for the desired symmetry and small dimensions required for patterns, I moved on. Once I learned to navigate the software at the ground level, I used ChatGPT to generate directions for a new user on how to make my own custom roll. First, Chat generated a method to map a random 2D geometric grayscale pattern onto a sample plane using noise nodes under the Geometry Nodes tab; this design was used as the repeat tile and was wrapped around a cylinder. Although this method was an improvement from the sculpting route in terms of consistent symmetry, there was still no way to have complete control over character design parameters. I continued to modify my method by engaging in a constant conversation with ChatGPT to clarify and provide feedback on how my roll appeared after certain steps.

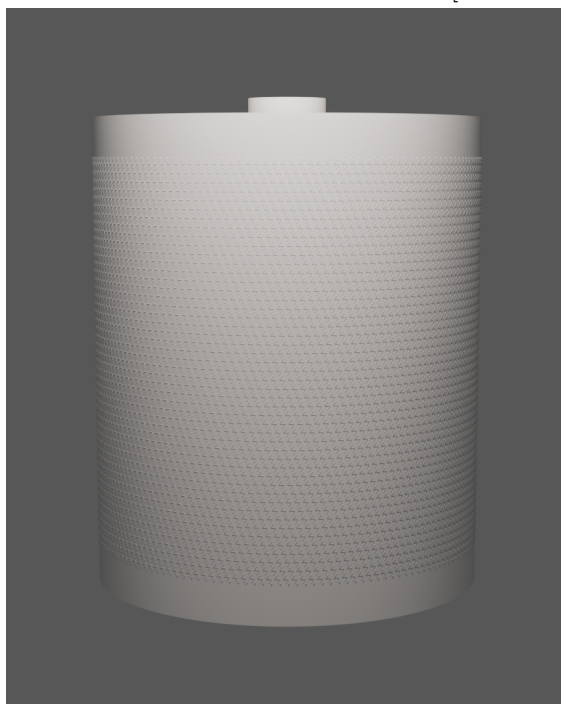
The final and most suitable of the methods was creating a base cylinder and applying a long chain of geometry nodes to produce and wrap a custom 3D texture sheet based on an AutoCAD sketch I created.



(a) Custom "lightning bolt" pattern sketch.



(b) Finished roll with 3D "lightning bolt" texture [zoomed in].



(c) Finished roll with 3D "lightning bolt" texture.

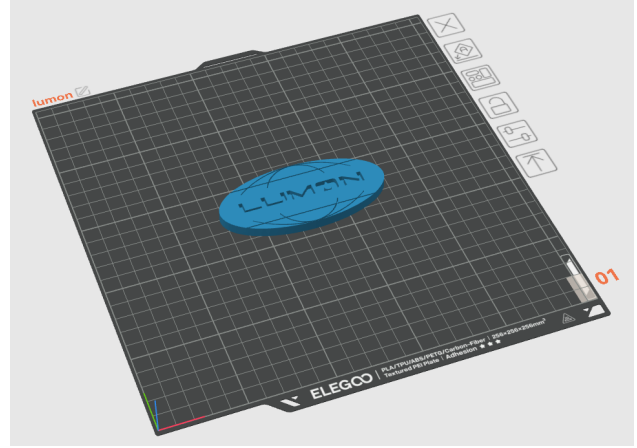
Figure 3: The initial AutoCAD sketch for my custom pattern and the final product created in Blender.

## 1.4 Inventor and 3D Printing

My introduction to CAD in high school was using Inventor Professional so I had a base familiarity with the program before the summer. By watching YouTube videos and receiving help from my mentor, I quickly reestablished my ability to use Inventor. I then learned how to slice a CAD model to prepare it for printing using the Elegoo Slicer for the Centauri Carbon printer. I created many different random models to practice my skills as demonstrated in Figure 4.



(a) Teacup and saucer model.



(b) Elegoo Slicer view of *Severance*'s Lumon logo.

Figure 4: Practice custom Inventor models.

## 1.5 Deflection Testing

### 1.5.1 Test Rig Creation Process

I started this process by reviewing the drawing of the flats and test setup from last year to familiarize myself with the test rig. Keeping the basis for the test rig the same, I altered some dimensions and angles to create a new flat drawing. I implemented the following changes: elongated the base cutouts to allow for an adjustable span of the sample tested, created cutouts for the deflection gauge stand in the main supports, and added in a couple of holes for more wood screws. This process familiarized me with AutoCAD as well as taught me the correct formatting and notation for drawing flats and bends. After finishing the flat drawings, my mentor taught me how to prepare my AutoCAD sketch using the Cincinnati Laser Programming and Nesting software for the Cincinnati CL920 Laser. I learned to prepare the sheet metal on the pallet and calibrate the laser to the gauge, type, and location of the metal and then run the program. The flats were then bent in a press brake by a coworker and the edges were deburred. The stainless steel pieces were screwed into a 0.5" thick piece of wood to ensure no unwanted movement occurred during testing.

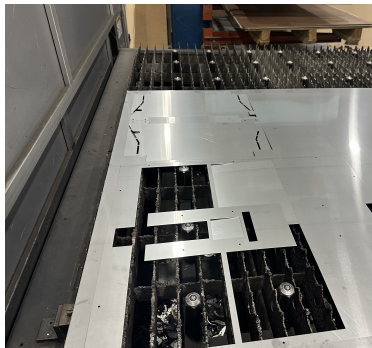


Figure 5: Rig Flats Fresh out the Laser Cutter.

### 1.5.2 Testing Sequence

A hole was drilled using a drill press as close to the center of each 6" x 6" sample for the connector hook. The sample was attached as depicted in Figure 6. Data was taken across three trials, at increments of 3, 6, 9, and 12 lbs for stainless steel and aluminum and 4, 8, 12, and 16 lbs for carbon. These twelve measurements were taken at each of four orientations: prime side up with the grain perpendicular, prime side up with the grain parallel, prime side down with the grain perpendicular, and prime side down with the grain parallel to the supports. The three data points for each mass increment were then averaged to get 16 final deflection numbers for each sample.

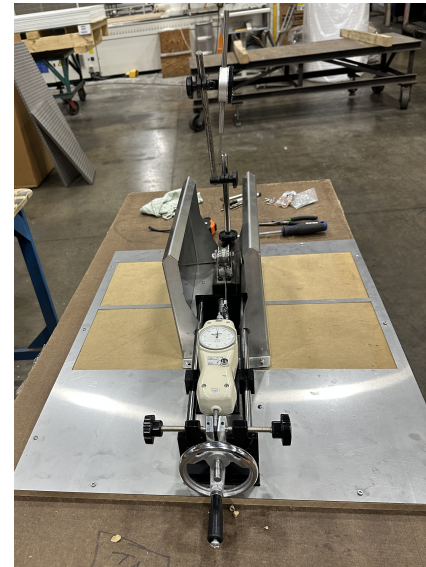
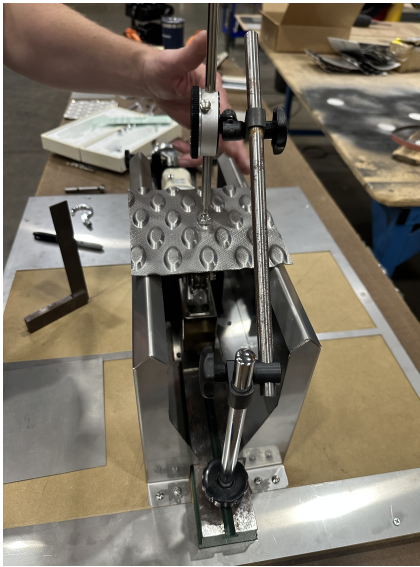


Figure 6: Deflection Test Rig.

## 2 Summary of Skills Developed

| Project              | Skill                                                                                |
|----------------------|--------------------------------------------------------------------------------------|
| Awesome Bump         | AwesomeBump software, applying metallic visuals for mock-up images                   |
| Sample Scanning      | Micrometers, optical profilometer scanning, profile and surface area data collection |
| 3D Modeling          | Inventor, Elegoo Slicer                                                              |
| Custom Roll Creation | AutoCAD, Blender, Inventor                                                           |
| Deflection Testing   | AutoCAD, Cincinnati laser software                                                   |